

How Estuaries Work: A Guide to Water Dynamics

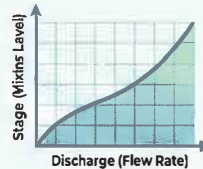
Estuaries are dynamic zones where freshwater from rivers meets and mixes with saltwater from the ocean. Understanding the volume and timing of these two water sources is critical for managing salinity, sediment transport, and the overall health of the ecosystem.



Measuring Freshwater Inflow

Freshwater inflow is a primary driver of estuarine conditions. It controls salinity, stratification, and the delivery of nutrients and pollutants.

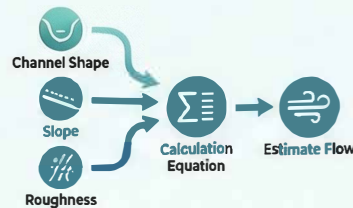
In Gauged Streams



Stage-Discharge Curve

A Stage-Discharge Curve provides continuous flow rate. This curve relates the measured water level (stage) to the flow rate (discharge).

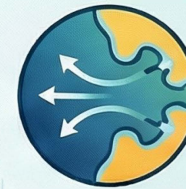
In Ungauged Streams



Manning's Equation estimates flow. This formula calculates discharge based on channel shape, slope, and roughness.

Understanding Tidal Exchange & Estuary Type

Tides create a cyclical flow of saltwater.



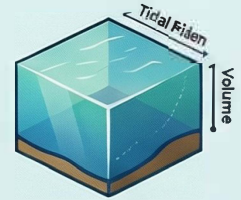
Ebb Tides

Ebb tides flow out to the sea.



Flood Tides

Flood tides flow into the estuary.



Tidal Prism Volume
It represents the total volume of water entering the estuary during one flood tide.

The Simons Ratio classifies estuaries by mixing type. It compares the volume of freshwater inflow to the tidal prism.



Simons Ratio & Estuary Type

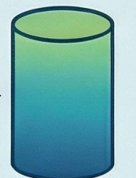


Stratified

Simons Ratio

> 1

< 0.1



Fully Mixed